Boudrouss Réda | CV

Software Engineer – OCaml, Rust, Static Analysis & WebAssembly – Paris

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Languages: French (Native), English (Professional), Arabic (Conversational)

Profile

Software engineer completing an MSc in Computer Science (Software Science & Technology) at Sorbonne University, specialized in **static analysis**, **abstract interpretation** and the design of **programming languages** and their runtimes. I ported **MOPSA** (an abstract-interpretation analyzer for C and Python, in OCaml/C/C++) to **WebAssembly** with a browser playground, and built an abstract-interpretation-based analyzer in **Rust**. At Hexaglobe I brought zero-coverage legacy products to ~3,000 tests with objective oracles and verification-gated AI agents. The common thread is building software people can trust.

Projects

Mopsa-wasm - github.com/rboudrouss/mopsa-emcc — 2026

· Ported **MOPSA**, an **abstract-interpretation** static analyzer for C and Python written in **OCaml**/C/C++, to **WebAssembly** by compiling the OCaml runtime and all native dependencies (Clang frontend, Apron, GMP, MPFR) with Emscripten and running MOPSA's OCaml bytecode on top.

· Worked on both sides of the **OCaml runtime / C FFI** boundary, generating a static table of ~1,400 C primitives (no `dlopen` on Wasm) and fixing an unsigned-wraparound bug in the runtime's `Tag_val` macro. Recompiled Apron's numerical domains to use exact GMP rationals, addressing the **soundness** hazard of Wasm's fixed rounding mode.

· Shipped a fully client-side playground and a technical write-up.

Reactant analyzer - github.com/rboudrouss/reactant-analyzer — 2026

· Designed and implemented a static analyzer based on **abstract interpretation**, written in **Rust**, targeting React web applications.

Education

MSc in Computer Science (Software Science & Technology) - Sorbonne University — 2024 – 2026

· Coursework: **abstract interpretation** and type systems (course led by Antoine Miné), inference of numerical properties, formal program verification, static typing, design & implementation of programming languages and runtimes (interpreters, compilers), **concurrent & distributed programming**.

BSc in Computer Science - Sorbonne University — 2020 – 2024

· Algorithms, computer architecture, operating systems, networks, databases, functional and object-oriented programming, with team projects in C, Java, OCaml, Python and TypeScript.

Experience

Quality Analyst Engineer (apprenticeship) - Hexaglobe (video streaming) — 2025 – 2026

· Designed complete **test strategies** (unit, integration, E2E) for zero-coverage legacy products, reaching ~**3,000 test cases** across 4 stacks (TypeScript, Python, PHP, JS).

· Industrialized verification-first **AI-agent test production**, combining sandboxed multi-agent orchestration, versioned guardrails, mandatory green tests and systematic human review to produce ~720 integration tests in 3 days.

· Built objective **quality oracles** (VMAF, black-frame/freeze detection, audio analysis) to validate a live video encoder, and **GitLab CI/CD** pipelines with shared components consumed by 4+ repositories.

Technical Lead, then President (2024–25) - Association Play Sorbonne Université — 2023 – 2026

· Designed and operated a virtualized **infrastructure** (Docker servers, CI/CD, networking) and built a **local-first** mobile web app. As president, led **100+ volunteers** for a cultural event welcoming 10,000+ visitors.

Skills

Languages: OCaml, Rust, TypeScript, Python, C, Java, SQL
Topics: static analysis & abstract interpretation, WebAssembly & Emscripten, concurrency & distributed systems, GitLab CI/CD, Docker